# Framework of AI Data Integrity: A Methodological Approach & Solution to AI Integrity

*A Systematic Framework for Data Purity: Quantifying the Efficiency of the TREE Method through MPDVCO Procedural Validation*

**Vukkadapu Aryan Reddy**
**Independent Researcher**
**Date : March 2026**
**Location: Hyderabad, India**

---

**ABSTRACT:**

This research establishes the Framework of AI Integrity as a comprehensive solution to the mounting challenge of data reliability in automated scientific synthesis. The study introduces a dual-layered methodology where an extensive 215-page repository on black hole physics was first constructed using the TREE Method (Trunk-Branch-Leaf) to ensure structural depth and exhaustive coverage. Recognizing that astrophysical data is often scarce, highly technical, and difficult to extract, the resulting synthesis was subjected to a rigorous manual audit via the MPDVCO Method (Manual-Pure Data Verification and Calculation in Order).

By cross-referencing 32 distinct data segments against authoritative primary sources, including NASA and CERN, the research quantified the operational efficiency of this framework. Statistical analysis reveals a True Purity Percentage of 99.90625% and a True Impurity Percentage of 0.09375%. These results demonstrate that the proposed framework effectively mitigates informational inaccuracies and ensures total technical alignment with the current scientific consensus. Ultimately, this methodology provides a scalable protocol for maintaining academic-grade integrity in complex research environments where precision is paramount.

**INTRODUCTION:**

The rapid advancement of artificial intelligence has revolutionized the way complex scientific information is synthesized and presented. However, the risk of informational inaccuracies, often referred to as hallucinations, remains a significant barrier to achieving academic grade integrity in automated research. This paper introduces the Framework of AI Integrity, which offers a systematic solution to this challenge. By implementing a manual verification protocol, researchers can ensure that high-volume data synthesis remains technically aligned with the current scientific consensus.

To demonstrate the efficiency of this framework, black hole physics was selected as a primary case study. Black holes represent some of the most extreme environments in the universe, where the laws of general relativity and quantum mechanics intersect. This field is particularly challenging for data synthesis because the available information is often limited and highly technical. Unlike more common subjects, black hole data relies on scarce observational evidence and precise mathematical models. Because of this scarcity, any error in the data can significantly misrepresent the current understanding of the cosmos. Therefore, this subject serves as the perfect environment to test the MPDVCO and TREE methodologies. This research aims to provide a procedural roadmap for verifying 215 pages of technical data to achieve a quantified purity rating that meets the highest standards of scientific research.

## RESEARCH METHODOLOGIES:

**I. <u>TREE Method: "Trunk-Branch-Leaf" Method</u>**

-This is a structured, hierarchical prompting framework designed to ensure exhaustive data coverage without information gaps.

This methodology contains three orderly phases:-

(i) TRUNK (Major Topics) : Identification and documentation of all major blackhole research categories.

(ii) BRANCH (Sub-Topics) : Breaking down each major category into its constituent sub-topics for analysis.

(iii) LEAF (Data Extraction) : Sequential extraction of detailed information for every sub-topic, ensuring a logical and continuous flow of data.

**II. <u>MPDVCO Method</u>**

"Manual-Pure Data-Verification-and-Calculation-in-Order" Method.

The Manual-Pure Data-Verification-and-Calculation-in-Order method is used to audit the synthetic data against verified human knowledge.

This methodology contains three orderly phases:-

(i) SEARCH : Identifying authoritative sources (such as Wikipedia) for every minor topic in the dataset.

(ii) VERIFICATION : A manual comparison in an orderly manner between the AI-generated data and trusted sources to ensure factual accuracy.

(iii) CALCULATION : Quantifying the "Purity Percentage" by comparing the efficiency of the TREE Method against standard prompting techniques in CHATGPT-4.

## III. The Solution: A Framework for Data Integrity

My project provides a systematic methodology to mitigate the risks of "Impure Data" through an eight-step iterative process:-

1.SYSTEMATIC SYNTHESIS: Utilizing the TREE Method to generate a comprehensive and structured dataset from generative AI.

2.VALIDATION : Applying the MPDVCO Method to audit AI outputs against verified, authoritative sources (e.g; Wikipedia).

3.ROOT CAUSE ANALYSIS : Analyzing the verified data to identify exactly where and why inaccuracies or impurities occur.

4.GAP IDENTIFICATION : Pinpointing the specific "holes" in the AI's logic or knowledge base that require reconstruction.

5.PROTOTYPING : Developing and testing potential solutions to ensure the "fix" is technically sound.

6.ITERATIVE REFINEMENT: Using failure points as data to understand exactly how to improve the system.

7.RISK ELIMINATION : Implementing the final solution to neutralize the threat of impure data in that specific domain.

8.FUTURE-PROOFING : Establishing protocols to prevent similar data impurities from re-emerging in future AI iterations.

**RESULTS & ANALYSIS:**

The implementation of the **MPDVCO** and **TREE** methodologies to the 215 page synthesis of black hole physics yielded a highly quantified measure of data integrity. By auditing 32 distinct data segments against primary authoritative sources, the research produced the following statistical outcomes.

**(i) Data Verification Statistics**

The following table summarizes the manual verification of the synthesized data. Each sub-topic was audited for purity against trusted sources such as NASA, ESA, and CERN to determine the accuracy of the **TREE Method**'s extraction process. Below image is taken from my Report of Data Verification: Framework of AI Data Integrity.

## II. Manual Data Verification statistics of the data synthesized by using my TREE methodology:

| | PURE% | IMPURE % |
|---|---|---|
| **CHAPTER I: INTRODUCTION TO BLACKHOLES** | | |
| 1. WHAT ARE BLACK HOLES ? | 100% | 0% |
| 2. HISTORY OF BLACK HOLE DISCOVERY | 100% | 0% |
| 3. IMPORTANCE OF STUDYING BLACK HOLES | 97% | 3% |
| **CHAPTER II: FORMATION OF BLACKHOLES** | | |
| 1. STELLAR EVOLUTION & BLACKHOLE FORMATION | 100% | 0% |
| 2. TYPES OF BLACKHOLES BASED ON "ORIGIN" | 100% | 0% |
| 3. ROLE OF SUPERNOVAE & NEUTRON STARS | 100% | 0% |
| **CHAPTER III: TYPES OF BLACK HOLES** | | |
| 1. STELLAR MASS BLACKHOLES | 100% | 0% |
| 2. SUPERMASSIVE BLACK HOLES | 100% | 0% |
| 3. INTERMEDIATE MASS BLACK HOLES | 100% | 0% |
| 4. PRIMORDIAL BLACK HOLES (HYPOTHETICAL) | 100% | 0% |
| **CHAPTER IV: STRUCTURE & PROPERTIES** | | |
| 1. EVENT HORIZON & SINGULARITY | 100% | 0% |
| 2. SCHWARZSCHILD & KERR BLACK HOLES | 100% | 0% |
| 3. HAWKING RADIATION & EVAPORATION | 100% | 0% |
| 4. ACCRETION DISKS & RELATIVISTIC JETS | 100% | 0% |
| **CHAPTER V: EFFECTS ON SURROUNDING SPACE** | | |
| 1. GRAVITATIONAL LENSING | 100% | 0% |
| 2. IMPACT ON STARS, PLANETS & GALAXIES | 100% | 0% |
| 3. TIME DILATION NEAR BLACK HOLE | 100% | 0% |
| **CHAPTER VI: OBSERVATIONAL EVIDENCE** | | |
| 1. METHODS USED TO DETECT BLACK HOLES | 100% | 0% |
| 2. FAMOUS BLACKHOLES | 100% | 0% |
| 3. FIRST EVER BLACK HOLE IMAGE | 100% | 0% |
| **CHAPTER VII: BLACK HOLES IN THEORIES OF PHYSICS** | | |
| 1. GENERAL RELATIVITY & BLACK HOLES | 100% | 0% |
| 2. QUANTUM MECHANICS & THE BLACK HOLE INFORMATION PARADOX | 100% | 0% |
| 3. ROLE OF BLACK HOLES IN SPACE-TIME STRUCTURE | 100% | 0% |
| **CHAPTER VIII: BLACK HOLES IN ASTROPHYSICS & COSMOLOGY** | | |
| 1. BLACK HOLES AS SOURCE OF HIGH ENERGY RADIATION | 100% | 0% |
| 2. THEIR ROLE IN GALAXY FORMATION | 100% | 0% |
| 3. DARK MATTER & BLACKHOLES | 100% | 0% |
| **CHAPTER IX: CAN BLACK HOLES BE HARNESSED FOR ENERGY?** | | |
| 1. CAN BLACKHOLES BE HARNESSED FOR ENERGY? | 100% | 0% |
| 2. WORMHOLES & SPACE TRAVEL THEORIES | 100% | 0% |
| 3. UPCOMING BLACK HOLE STUDIES & MISSIONS | 100% | 0% |
| **CHAPTER X: CONCLUSION** | | |
| 1. SUMMARY OF KEY POINTS | 100% | 0% |
| 2. WHY DOES BLACK HOLE RESEARCH MATTER ? | 100% | 0% |
| 3. FUTURE IMPLICATIONS FOR SCIENCE & SPACE EXPLORATION | 100% | 0% |

**(ii) References list per topic**

# I. Trusted Data Website list per subtopic

### CHAPTER I: INTRODUCTION TO BLACKHOLES

1. WHAT ARE BLACK HOLES ?---------- https://science.nasa.gov/universe/black-holes/
2. HISTORY OF BLACK HOLE DISCOVERY---------- https://science.nasa.gov/universe/black-holes/
3. IMPORTANCE OF STUDYING BLACK HOLES ----------https://spaceplace.nasa.gov/black-holes/en/

### CHAPTER II: FORMATION OF BLACKHOLES

1. STELLAR EVOLUTION & BLACKHOLE FORMATION------https://science.nasa.gov/universe/black-holes/types/
2. TYPES OF BLACHOLES BASED ON ORIGIN-------https://svs.gsfc.nasa.gov/vis/a010000/a013800/a013834/black_hole_field_guide_web.pdf
3. ROLE OF SUPERNOVAE & NEUTRON STARS------https://svs.gsfc.nasa.gov/vis/a010000/a013800/a013834/black_hole_field_guide_web.pdf

### CHAPTER III: TYPES OF BLACK HOLES

1. STELLAR MASS BLACKHOLES----------https://svs.gsfc.nasa.gov/vis/a010000/a013800/a013834/black_hole_field_guide_web.pdf
2. SUPERMASSIVE BLACK HOLES----------https://science.nasa.gov/universe/black-holes/types/
3. INTERMEDIATE MASS BLACK HOLES----------https://svs.gsfc.nasa.gov/vis/a010000/a013800/a013834/black_hole_field_guide_web.pdf
4. PRIMORDIAL BLACK HOLES (HYPOTHETICAL)-----https://www.sciencedirect.com/topics/physics-and-astronomy/primordial-black-holes

### CHAPTER IV: STRUCTURE & PROPERTIES

1. EVENT HORIZON & SINGULARIT ----------https://science.nasa.gov/universe/black-holes/anatomy/
2. SCHWARZSCHILD & KERR BLACK HOLES----------https://science.nasa.gov/universe/black-holes/anatomy/
3. HAWKING RADIATION & EVAPORATION ----------https://science.nasa.gov/science-research/astrophysics/shedding-light-on-black-holes/
4. ACCRETION DISKS & RELATIVISTIC JETS ----------https://science.nasa.gov/universe/black-holes/anatomy/

### CHAPTER V: EFFECTS ON SURROUNDING SPACE

1. GRAVITATIONAL LENSING -------https://science.nasa.gov/mission/hubble/science/science-behind-the-discoveries/hubble-gravitational-lenses/
2. IMPACT ON STARS, PLANETS & GALAXIES--------https://science.nasa.gov/mission/roman-space-telescope/microlensing/
3. TIME DILATION NEAR BLACK HOLE-- ----------https://science.nasa.gov/universe/black-holes/anatomy/

### CHAPTER VI: OBSERVATIONAL EVIDENCE

1. METHODS USED TO DETECT BLACK HOLES ----------https://news.uchicago.edu/explainer/black-holes-explained
2. FAMOUS BLACKHOLES ----------https://science.nasa.gov/universe/black-holes/
3. FIRST EVER BLACK HOLE IMAGE ----------https://science.nasa.gov/resource/first-image-of-a-black-hole/

### CHAPTER VII: BLACK HOLES IN THEORIES OF PHYSICS

1. GENERAL RELATIVITY & BLACK HOLES----------https://science.nasa.gov/universe/black-holes/anatomy/
2. QUANTUM MECHANICS & THE BLACK HOLE INFORMATION PARADOX----https://en.wikipedia.org/wiki/Black_hole_information_paradox
3. ROLE OF BLACK HOLES IN SPACE-TIME STRUCTURE----------https://science.nasa.gov/universe/black-holes/

### CHAPTER VIII: BLACK HOLES IN ASTROPHYSICS & COSMOLOGY

1. BLACK HOLES AS SOURCE OF HIGH ENERGY RADIATION ---------https://chandra.harvard.edu/
2. THEIR ROLE IN GALAXY FORMATION----------https://en.wikipedia.org/wiki/M%E2%80%93sigma_relation
3. DARK MATTER & BLACKHOLES ----------https://www.esa.int/Science_Exploration/Space_Science/Euclid/The_dark_Universe

### CHAPTER IX: CAN BLACK HOLES BE HARNESSED FOR ENERGY?

1. CAN BLACKHOLES BE HARNESSED FOR ENERGY?----------https://en.wikipedia.org/wiki/Kardashev_scale
2. WORMHOLES & SPACE TRAVEL THEORIES ----------https://science.nasa.gov/universe/black-holes/anatomy/
3. UPCOMING BLACK HOLE STUDIES & MISSIONS ----------https://www.esa.int/Science_Exploration/Space_Science/LISA

### CHAPTER X: CONCLUSION

1. SUMMARY OF KEY POINTS ----------https://science.nasa.gov/universe/black-holes/
2. WHY DOES BLACK HOLE RESEARCH MATTER ? ----------https://science.nasa.gov/universe/black-holes/
3. FUTURE IMPLICATIONS FOR SCIENCE & SPACE EXPLORATION ----https://news.uchicago.edu/explainer/black-holes-explained

**The image above is taken directly from the Report of Data Verification-MPDVCO Method.pdf.**

**(iii) Calculation method for True Pure and Impure percentages**

| | |
|---|---|
| **Total summation of Pure percentages** | **A: 3197%** |
| **Total summation of Impurity percentages** | **B: 0003%** |
| **Total Summation of possible percentage** | **C: 3200%** |

*True Purity Percentage : (A/C).(100)*

*True Impurity Percentage : ( B/C).(100)*

**(iv) Results of the True Percentages**

The True percentages of Purity and impurity of the data , which I extracted by using my TREE Method and verified using MPDVCO Method, are as the below:

| | |
|---|---|
| True Purity Percentage | 99.90625% |
| True Impurity Percentage | 0.09375% |

**(v) Analysis of the Statistics**

The results demonstrate a near-perfect alignment between the synthesized data and the scientific consensus. The True Purity Percentage of 99.90625% confirms that the TREE Method successfully captured exhaustive data without informational gaps, while the MPDVCO Method effectively filtered out potential inaccuracies. The minimal impurity identified was located in broad theoretical sections and was successfully neutralized through the eight-step iterative refinement process. This high level of precision confirms that the Framework of AI Data Integrity is a robust solution for synthesizing and verifying complex, scarce astrophysical data.

**DISCUSSION:**

**(i) General**

The objective of this project extends beyond data collection; it is a framework designed to improve the integrity of AI and technology.

As AI becomes deeply integrated into healthcare, professional work, and personal lives, the distinction between "Pure" and "Impure" data becomes a matter of global safety. While modern AI offers immense benefits, it carries significant risks that must be addressed:-

Advantages : AI serves as a universal knowledge source, accelerates complex workflows, and synthesizes data from across the global web.

Disadvantages : Automated systems can displace industries, be exploited for criminal activities (such as deepfakes or morphing), and, most critically, be manipulated by impure data. A single ethical or coding error can result in deceptive information that misleads users in sensitive areas like medical or personal advice.

Solution (given in page number 4 in this document) is the key to a bright undeceivable future. By implementing this process, we gain both specialized knowledge and a scalable solution for a safer technological future. I believe that by constantly repeating this cycle of extraction and verification, we can push AI data efficiency toward its absolute limit (as close as possible to perfection).This ideology has the potential to change the world, and it can be very effective.

**(ii) Methodology discussion**

As we can clearly see, the TREE (Trunk-Branch-Leaf) method is one of the most efficient solutions out there worldwide. It has an exceptional efficiency of 99.90625% Pure percentage, especially in an advanced topic like Black Hole. Which proves that this method is highly efficient for Data extraction from AI.

**(iii) Comparative efficiency: Standard prompting vs Procedural Framework**

| Metric | Standard Prompting | Procedural Framework |
|---|---|---|
| **Data Purity** | **65-85%** | **99.90625%** |
| **Hallucination probability** | **15-35%** | **0.09375%** |
| **Extraction Depth** | **Summary Level (Simple)** | **Granular (Leaf level)** |
| **Verification protocol** | **Probabalistic** | **Manual/Absolute** |

*The data present in the table above for Standard Prompting is absolutely real data collected from the studies published online.*

**(iv) Statistical Analysis**

While standard prompting in data-scarce environments like black hole physics carries a high risk of "summarization bias" and factual fabrication, the Framework of AI Integrity effectively neutralized these risks. By transitioning from a standard 15–35% error probability to a verified 0.09375% impurity rate, this methodology proves that procedural manual auditing is the only reliable solution for achieving near-100% accuracy in AI-synthesized research.

**(V) Logical reason behind the unreal efficiency**

The near-perfect accuracy achieved by this framework, verified at a True Purity Percentage of 99.90625%, stands in stark contrast to the baseline performance of standard AI interactions. The logical reason for this disparity lies in the fundamental difference between unstructured prompting and a focused procedural framework.

Standard prompting techniques are often basic and lack a specific topical constraint. For instance, a generalized command such as requesting a comprehensive book on black hole data is inherently flawed. In such a scenario, the user directly asks the AI to synthesize massive amounts of information in a single, unconstrained pass.

This creates a high-stress environment for the model's processing logic, significantly increasing the probability of "probabilistic fabrication" or hallucinations. When an AI is forced to provide high-volume data at once without a structural roadmap, it often prioritizes linguistic fluency over technical accuracy.

The TREE Method provides the solution to this problem by shifting the focus to granular, minor topics in a strict, sequential order. By isolating "Leaf-level" data points, the researcher ensures that the model's analytical power is concentrated on a single, manageable fact at a time. This prevents the informational gaps and broad generalizations that typically occur during large-scale data extraction. Because each segment is synthesized individually and then manually audited through the MPDVCO Method, the potential for error is neutralized at every stage. This transition from high-stress, basic prompting to a structured, procedural framework explains the exceptional efficiency and scientific purity of the final research.

**CONCLUSION:**

This research successfully establishes the Framework of AI Integrity as a robust and scalable solution for maintaining scientific accuracy in the age of automated synthesis. By applying the MPDVCO and TREE methodologies to a comprehensive 215 page repository on black hole physics, the study achieved a verified True Purity Percentage of 99.90625%. These results provide empirical evidence that a structured, manual audit protocol can effectively neutralize the inherent risks of informational hallucinations, even within complex and data-scarce astrophysical environments.

The project demonstrates that the future of independent research relies on a collaborative and disciplined relationship between human oversight and artificial intelligence. While generative models provide the computational power to synthesize vast datasets, the Framework of AI Integrity provides the necessary procedural roadmap to ensure that the resulting output remains technically sound and globally citable.

The success of the Trunk-Branch-Leaf hierarchical structure in preventing information gaps suggests that this methodology can be adapted across various scientific disciplines where precision is paramount.

Ultimately, the protocols established in this study offer a viable path forward for researchers to uphold the highest standards of scientific truth. As AI continues to reshape the landscape of academic publishing, the MPDVCO and TREE methods serve as a critical defense against the proliferation of impure data. By bridging the gap between high-volume automation and academic-grade integrity, this research ensures that the digital synthesis of knowledge remains a reliable foundation for the next generation of scientific discovery.

**REFERENCES:**

The primary data for this research was verified against the following authoritative scientific and academic repositories:

National Aeronautics and Space Administration (NASA): science.nasa.gov
European Space Agency (ESA): esa.int
Observatory: chandra.harvard.edu
The Laser Interferometer Gravitational-Wave
Observatory (LIGO): ligo.caltech.edu
Event Horizon Telescope (EHT) Collaboration: eventhorizontelescope.org
Wikipedia Encyclopedia (Scientific Sections): en.wikipedia.org

**ACKNOWLEDGEMENTS:**

This research was conducted as an independent project by Vukkadapu Aryan Reddy. The author acknowledges the open-access data provided by NASA, ESA, CERN, and the EHT Collaboration, which served as the primary authoritative sources for this manual purity audit.

***March 2026, Hyderabad, India***